

$$\begin{aligned} & \frac{(2)}{\phi D} \rightarrow \frac{1}{16\pi^2} \left(\frac{1}{9} g_2^4 \text{LF}_{3,0}[m_2] + \frac{1}{6} g_2^4 \text{LF}_{4,-1}[m_2] - \frac{8}{45} g_2^4 \text{LF}_{5,-2}[m_2] - \right. \\ & \frac{1}{27} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0}[m_{\mathbf{d}}^{\mathbf{p}}] + \frac{5}{72} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1}[m_{\mathbf{d}}^{\mathbf{p}}] - \frac{4}{135} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2}[m_{\mathbf{d}}^{\mathbf{p}}] - \frac{1}{9} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0}[m_{\mathbf{e}}^{\mathbf{p}}] + \\ & \frac{5}{24} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1}[m_{\mathbf{e}}^{\mathbf{p}}] - \frac{4}{45} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2}[m_{\mathbf{e}}^{\mathbf{p}}] + \frac{1}{18} \sum_{\mathbf{p}} (-g_1^4 + g_2^4) \text{LF}_{3,0}[m_{\mathbf{l}}^{\mathbf{p}}] + \\ & \frac{5}{48} \sum_{\mathbf{p}} (g_1^4 - g_2^4) \text{LF}_{4,-1}[m_{\mathbf{l}}^{\mathbf{p}}] - \frac{2}{45} \sum_{\mathbf{p}} (g_1^4 - g_2^4) \text{LF}_{5,-2}[m_{\mathbf{l}}^{\mathbf{p}}] - \frac{1}{54} \sum_{\mathbf{p}} (g_1^4 - 9g_2^4) \text{LF}_{3,0}[m_{\mathbf{q}}^{\mathbf{p}}] + \\ & \frac{5}{144} \sum_{\mathbf{p}} (g_1^4 - 9g_2^4) \text{LF}_{4,-1}[m_{\mathbf{q}}^{\mathbf{p}}] - \frac{2}{135} \sum_{\mathbf{p}} (g_1^4 - 9g_2^4) \text{LF}_{5,-2}[m_{\mathbf{q}}^{\mathbf{p}}] - \\ & \frac{4}{27} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0}[m_{\mathbf{u}}^{\mathbf{p}}] + \frac{5}{18} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1}[m_{\mathbf{u}}^{\mathbf{p}}] - \frac{16}{135} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2}[m_{\mathbf{u}}^{\mathbf{p}}] + \\ & \frac{1}{18} (-g_1^4 + g_2^4) \text{LF}_{3,0}[\tilde{\mu}] + \frac{1}{12} (-g_1^4 + g_2^4) \text{LF}_{4,-1}[\tilde{\mu}] + \frac{4}{45} (g_1^4 - g_2^4) \text{LF}_{5,-2}[\tilde{\mu}] + \\ & \frac{1}{16} g_1^4 \text{LF}_{2,2,-1}[m_1, \tilde{\mu}] + \frac{1}{8} g_1^4 m_1^2 \text{LF}_{2,2,0}[m_1, \tilde{\mu}] - \frac{2}{3} g_2^4 \text{LF}_{2,1,0}[m_2, \tilde{\mu}] - \\ & \frac{17}{48} g_2^4 \text{LF}_{2,2,-1}[m_2, \tilde{\mu}] + \frac{1}{8} g_2^4 m_2^2 \text{LF}_{2,2,0}[m_2, \tilde{\mu}] + \frac{1}{3} g_2^4 \text{LF}_{3,1,-1}[m_2, \tilde{\mu}] + \\ & \frac{1}{3} g_2^4 \text{LF}_{3,2,-2}[m_2, \tilde{\mu}] - \frac{1}{6} g_1^2 (\overline{a_d^{\text{pr}}} a_d^{\text{pr}} + \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}}) \text{LF}_{2,2,0}[m_{\mathbf{d}}^{\text{r}}, m_{\mathbf{q}}^{\text{p}}] + \\ & \frac{1}{12} g_1^2 (\overline{a_d^{\text{pr}}} a_d^{\text{pr}} + \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}}) \text{LF}_{3,2,-1}[m_{\mathbf{d}}^{\text{r}}, m_{\mathbf{q}}^{\text{p}}] - \frac{1}{6} g_1^2 (\overline{a_e^{\text{pr}}} a_e^{\text{pr}} + \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}}) \\ & \text{LF}_{2,2,0}[m_{\mathbf{e}}^{\text{r}}, m_{\mathbf{l}}^{\text{p}}] + \frac{1}{12} g_1^2 (\overline{a_e^{\text{pr}}} a_e^{\text{pr}} + \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}}) \text{LF}_{3,2,-1}[m_{\mathbf{e}}^{\text{r}}, m_{\mathbf{l}}^{\text{p}}] + \\ & \frac{1}{6} g_1^2 (\overline{a_e^{\text{pr}}} a_e^{\text{pr}} + \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}}) \text{LF}_{3,1,0}[m_{\mathbf{l}}^{\text{p}}, m_{\mathbf{e}}^{\text{r}}] + \frac{1}{6} g_1^2 (\overline{a_e^{\text{pr}}} a_e^{\text{pr}} + \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}}) \\ & \text{LF}_{3,2,-1}[m_{\mathbf{l}}^{\text{p}}, m_{\mathbf{e}}^{\text{r}}] - \frac{1}{8} (3g_1^2 - g_2^2) (\overline{a_e^{\text{pr}}} a_e^{\text{pr}} + \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}}) \text{LF}_{4,1,-1}[m_{\mathbf{l}}^{\text{p}}, m_{\mathbf{e}}^{\text{r}}] + \\ & \frac{1}{6} (g_1^2 - g_2^2) (\overline{a_e^{\text{pr}}} a_e^{\text{pr}} + \tilde{\mu}^2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}}) \text{LF}_{5,1,-2}[m_{\mathbf{l}}^{\text{p}}, m_{\mathbf{e}}^{\text{r}}] + \frac{1}{6} g_1^2 (\overline{a_d^{\text{pr}}} a_d^{\text{pr}} + \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}}) \\ & \text{LF}_{3,1,0}[m_{\mathbf{q}}^{\text{p}}, m_{\mathbf{d}}^{\text{r}}] + \frac{1}{6} g_1^2 (\overline{a_d^{\text{pr}}} a_d^{\text{pr}} + \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}}) \text{LF}_{3,2,-1}[m_{\mathbf{q}}^{\text{p}}, m_{\mathbf{d}}^{\text{r}}] - \\ & \frac{1}{8} (5g_1^2 - 3g_2^2) (\overline{a_d^{\text{pr}}} a_d^{\text{pr}} + \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}}) \text{LF}_{4,1,-1}[m_{\mathbf{q}}^{\text{p}}, m_{\mathbf{d}}^{\text{r}}] + \\ & \frac{1}{2} (g_1^2 - g_2^2) (\overline{a_d^{\text{pr}}} a_d^{\text{pr}} + \tilde{\mu}^2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}}) \text{LF}_{5,1,-2}[m_{\mathbf{q}}^{\text{p}}, m_{\mathbf{d}}^{\text{r}}] + \\ & \frac{1}{12} (g_1^2 + 3g_2^2) (\overline{a_u^{\text{pr}}} a_u^{\text{pr}} + \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}}) \text{LF}_{2,2,0}[m_{\mathbf{q}}^{\text{p}}, m_{\mathbf{u}}^{\text{r}}] - \\ & \frac{1}{24} (g_1^2 + 3g_2^2) (\overline{a_u^{\text{pr}}} a_u^{\text{pr}} + \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}}) \text{LF}_{3,2,-1}[m_{\mathbf{q}}^{\text{p}}, m_{\mathbf{u}}^{\text{r}}] - \\ & \frac{1}{12} (g_1^2 + 3g_2^2) (\overline{a_u^{\text{pr}}} a_u^{\text{pr}} + \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}}) \text{LF}_{3,1,0}[m_{\mathbf{u}}^{\text{r}}, m_{\mathbf{q}}^{\text{p}}] - \\ & \frac{1}{12} (g_1^2 + 3g_2^2) (\overline{a_u^{\text{pr}}} a_u^{\text{pr}} + \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}}) \text{LF}_{3,2,-1}[m_{\mathbf{u}}^{\text{r}}, m_{\mathbf{q}}^{\text{p}}] - \\ & \frac{1}{4} (g_1^2 - 3g_2^2) (\overline{a_u^{\text{pr}}} a_u^{\text{pr}} + \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}}) \text{LF}_{4,1,-1}[m_{\mathbf{u}}^{\text{r}}, m_{\mathbf{q}}^{\text{p}}] + \\ & \frac{1}{2} (g_1^2 - g_2^2) (\overline{a_u^{\text{pr}}} a_u^{\text{pr}} + \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}}) \text{LF}_{5,1,-2}[m_{\mathbf{u}}^{\text{r}}, m_{\mathbf{q}}^{\text{p}}] + \\ & \frac{3}{2} \overline{y_d^{\text{pr}}} y_d^{\text{sr}} \overline{y_u^{\text{st}}} y_u^{\text{pt}} \text{LF}_{2,1,0}[m_{\mathbf{u}}^{\text{t}}, m_{\mathbf{d}}^{\text{r}}] - \frac{3}{2} \overline{y_d^{\text{pr}}} y_d^{\text{sr}} \overline{y_u^{\text{st}}} y_u^{\text{pt}} \text{LF}_{3,1,-1}[m_{\mathbf{u}}^{\text{t}}, m_{\mathbf{d}}^{\text{r}}] + \\ & \frac{1}{8} g_1^4 \text{LF}_{3,2,-2}[\tilde{\mu}, m_1] - \frac{1}{8} g_1^4 m_1^2 \text{LF}_{3,2,-1}[\tilde{\mu}, m_1] + \frac{5}{12} (g_1^4 - g_1^2 g_2^2) \text{LF}_{4,1,-2}[\tilde{\mu}, m_1] - \\ & \frac{1}{8} g_1^4 \text{LF}_{4,2,-3}[\tilde{\mu}, m_1] + \frac{1}{8} g_1^4 m_1^2 \text{LF}_{4,2,-2}[\tilde{\mu}, m_1] + \frac{1}{3} (-g_1^4 + g_1^2 g_2^2) \text{LF}_{5,1,-3}[\tilde{\mu}, m_1] + \\ & \frac{4}{3} g_2^4 \text{LF}_{3,1,-1}[\tilde{\mu}, m_2] + \frac{31}{24} g_2^4 \text{LF}_{3,2,-2}[\tilde{\mu}, m_2] - \frac{1}{8} g_2^4 m_2^2 \text{LF}_{3,2,-1}[\tilde{\mu}, m_2] + \\ & \frac{1}{4} g_2^2 (5g_1^2 - 9g_2^2) \text{LF}_{4,1,-2}[\tilde{\mu}, m_2] - \frac{5}{8} g_2^4 \text{LF}_{4,2,-3}[\tilde{\mu}, m_2] + \frac{1}{8} g_2^4 (m_2^2 + 4\tilde{\mu}^2) \text{LF}_{4,2,-2}[\tilde{\mu}, m_2] + \\ & (-g_1^2 g_2^2 + g_2^4) \text{LF}_{5,1,-3}[\tilde{\mu}, m_2] + 3g_1^2 g_2^2 \tilde{\mu}^2 \text{LF}_{2,2,1,-1}[m_2, \tilde{\mu}, m_1] - \\ & 2g_1^2 g_2^2 \tilde{\mu}^2$$